

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0550

Sugar Hill 138 kV
May 18, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0550. The Interconnection Customer proposes a 343.7 MW wind facility interconnecting at the Sugar Hill 138 kV substation in the Pennsylvania area. The request was received on December 09, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 248 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,550,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Blackrock to Eklutna 345kV
Contingency: Caldwell to Piney Creek Project 345kV
Loading: 105.7% (Rating: 1,136 MVA)
DFAX: 5.9%
Pre-project loading: 103.9%
2. Facility: Yucca to Regal 138kV
Contingency: Allegheny Energy Units 3 4 & 5 to Southeast Vicksburg 500kV
Loading: 101.5% (Rating: 211 MVA)
DFAX: 10.7%
Pre-project loading: 84.1%
3. Facility: Maricao 6301 to Sta J 138kV line
Contingency: Regal to Yucca 138kV
Loading: 100.2% (Rating: 159 MVA)
DFAX: 11.2%
Pre-project loading: 76.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Blackrock to Eklutna 345kV	Reconductor line	\$20,023,000	33% (\$6,681,000)
Yucca to Regal 138kV	Reconductor line	\$539,000	79% (\$427,000)
Maricao 6301 to Sta J 138kV line	Reconductor line	\$442,000	100% (\$442,000)
	Total allocated to project		\$7,550,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.